

VOICE BASED CONCEPT UNDERSTANDING ANALYSER

Project Description:

The Voice-Based Concept Understanding Analyser is an AI-powered academic evaluation system designed to automatically analyze and score a student's spoken explanation of technical concepts. The core objective is to replace manual transcription and subjective evaluation with automated, multi-dimensional speech analysis.

The system allows users to upload an audio recording (e.g., an MP4 or WAV file containing speech) along with the student's name. It automatically performs Speech-to-Text conversion using Google's Speech Recognition engine and extracts key audio and linguistic features. The processed text is then evaluated using Google's Gemini 2.5-flash model via the google-genai SDK to dynamically assess conceptual understanding, generate qualitative feedback, and track fluency metrics like filler words (um, uh, like, ah). Finally, it generates comprehensive performance analytics, displays an interactive audio waveform graph, tracks session records in a local database, and provides a downloadable PDF evaluation report.

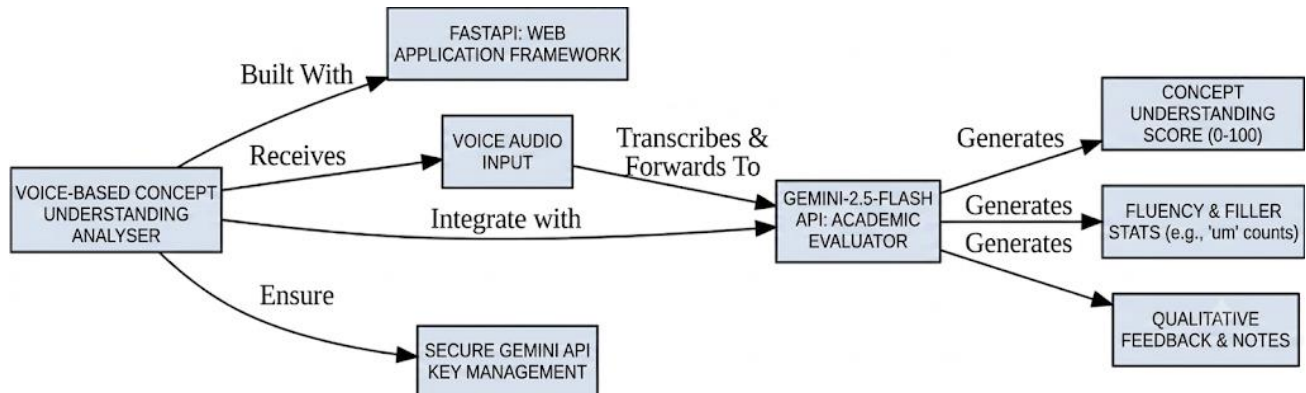
Scenarios

Scenario 1: A student uploads an explanation of "Machine Learning" and selects an Academic evaluation. The system generates an exact textual transcription of the audio, calculates physical metrics like a 27.82-second duration and 160 WPM pacing, renders a visual waveform graph, and outputs a "Strong Understanding" grade along with a comprehensive PDF report in seconds

Scenario 2: An interviewer evaluates a candidate describing "Cloud Computing architectures." The FastAPI backend processes the uploaded audio stream, handles data storage into the SQLite database, and queries the Gemini 2.5-flash model. The AI dynamically predicts a precise counts of speech flaws—such as detecting 0 "ums" or "ahs"—and generates a structural qualitative feedback sentence focusing on technical fluency.

Scenario 3: A student practicing for a technical presentation uploads a weak, fragmented explanation of a concept. The transcription engine converts the speech to text, while the AI integration tier catches conceptual gaps, flagging the performance under a "Poor Understanding" or "Moderate Understanding" level. The dashboard instantly updates the student's metrics table showing low semantic similarity and provides a one-click downloadable PDF mapping out areas for correction.

Technical Architecture:



Description: The Voice-Based Concept Understanding Analyser utilizes a modular three-tier architecture to deliver rapid, AI-driven speech evaluation. The frontend provides a responsive user interface built with interactive visualization layouts, while the FastAPI-based backend orchestrates audio processing, metric calculation, and prompt construction. It interfaces with the Google Generative AI API using the Gemini 2.5-flash model to generate tailored score breakdowns, filler word stats, and qualitative feedback, with secure database logging managed locally via SQLite.

Pre-requisites:

- Python Programming Knowledge
- FastAPI Framework Knowledge
- Streamlit Framework Knowledge
- Natural Language Processing Concepts
- Generative AI and LLM Architectures
- Audio Processing and Speech Recognition Fundamentals
- REST API Concepts
- Relational Database Concepts (SQLite)
- Git and GitHub Basics

Project Workflow:

Milestone 1: Model Selection and Architecture

- **Activity 1.1:** Generate a Google Gemini API key and configure it securely inside the project environment variables (.env).
- **Activity 1.2:** Select the gemini-2.5-flash model via the google-genai SDK for low-latency text evaluation and structured JSON extraction.
- **Activity 1.3:** Define the three-tier application architecture, detailing interactions between the interactive UI, FastAPI backend, and Gemini API.
- **Activity 1.4:** Set up the development environment, initializing the local SQLite database file and installing dependencies from requirements.txt.

Milestone 2: Core Functionalities Development

- **Activity 2.1:** Develop core pipelines: Speech-to-Text transcription, audio waveform image rendering, and WPM pace tracking.
- **Activity 2.2:** Design structural academic prompts using strict JSON schemas to fetch quantitative scores, qualitative notes, and fluency filler stats.

Milestone 3: Main.py Development

- **Activity 3.1:** Write the main application logic in main.py, establishing REST API endpoints (/upload-audio/, /api/records, /download-pdf/{record_id}) and managing local CRUD operations with sqlite3.

Milestone 4: Frontend Development

- **Activity 4.1:** Build an interactive user interface dashboard utilizing layout frameworks to handle student data inputs and display visualization charts.
- **Activity 4.2:** Construct dynamic result elements to render audio waveforms, score progress rings, filler word counts, and one-click PDF download options.

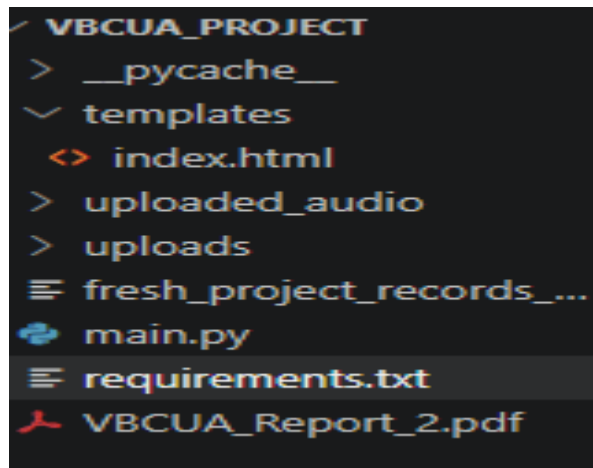
Milestone 5: Deployment

- **Activity 5.1:** Prepare the application for local hosting by executing Uvicorn server parameters, ensuring correct system paths for FFmpeg audio handling.
- **Activity 5.2:** Test and validate local database logging, file system uploads, and streaming PDF responses under concurrent endpoint triggers.

Milestone 6: Conclusion

- **Activity 6.1:** Deliver a fully automated, functional Voice-Based Concept Understanding Analyser that successfully eliminates manual oral grading overhead.

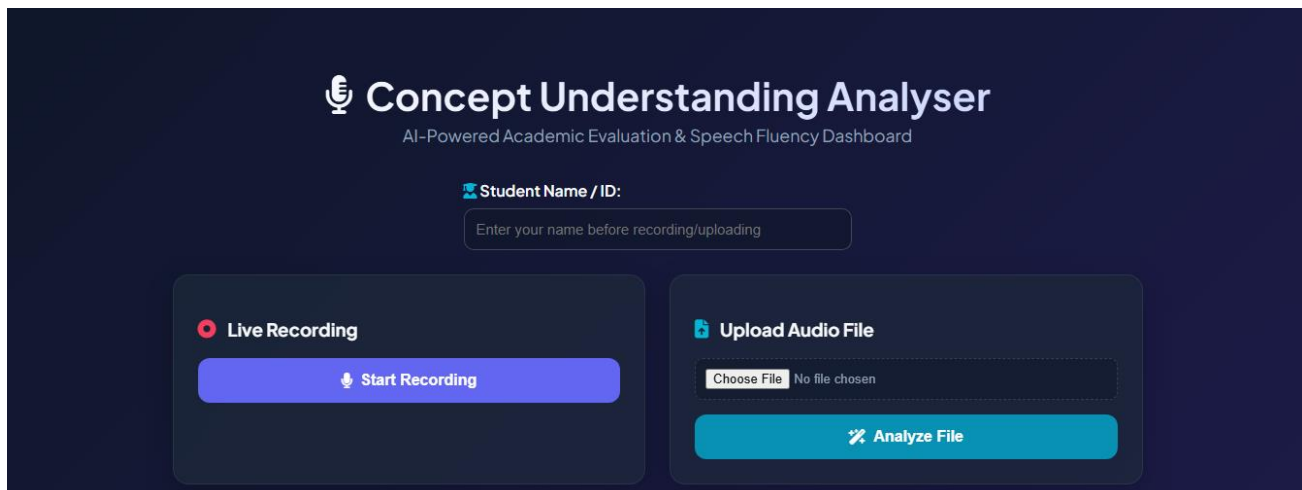
1. Project Structure



2. Running the Application

```
PS C:\Users\MANEEJA VAGU\Downloads\vbcua_project> uvicorn main:app --reload
INFO: Will watch for changes in these directories: ['C:\\Users\\MANEEJA VAGU\\Downloads\\vbcua_project']
INFO: Uvicorn running on http://127.0.0.1:8000 (Press CTRL+C to quit)
INFO: Started reloader process [16800] using StatReload
```

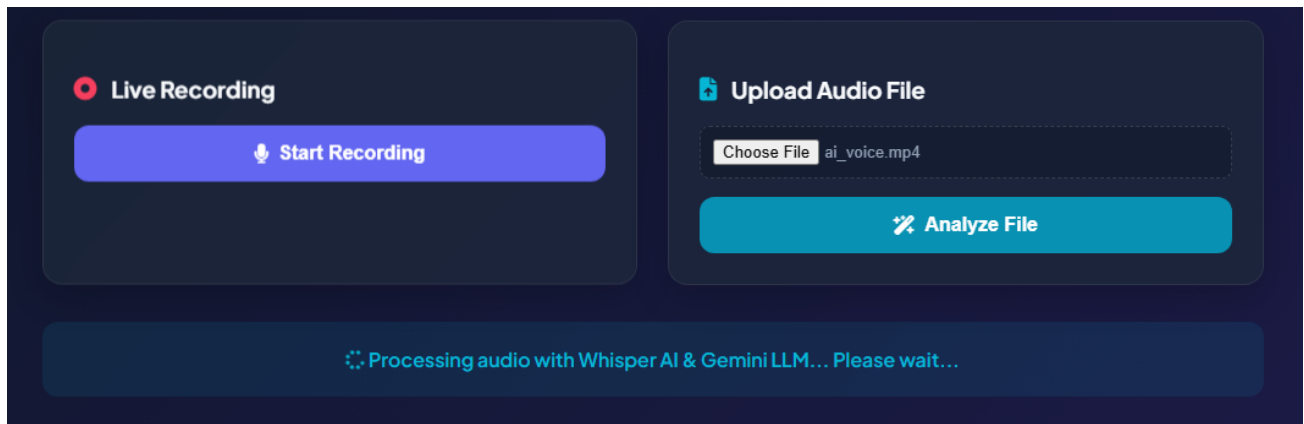
3. Home Dashboard / User Input Interface



The screenshot shows the 'Concept Understanding Analyser' web interface. It features a dark blue background with a white microphone icon and the title 'Concept Understanding Analyser' in white. Below the title is the subtitle 'AI-Powered Academic Evaluation & Speech Fluency Dashboard'. A form for 'Student Name / ID:' is present, with a placeholder text 'Enter your name before recording/uploading'. Below this, there are two main sections: 'Live Recording' and 'Upload Audio File'. The 'Live Recording' section has a red circular icon and a blue button labeled 'Start Recording'. The 'Upload Audio File' section has a blue square icon, a file input field with 'Choose File' and 'No file chosen' options, and a blue button labeled 'Analyze File'.

This is the primary user interface built with HTML/Streamlit layouts where the evaluator or student inputs the student's name and uploads the voice recording file (MP4 or WAV format).

4. Audio Processing:



The interface is divided into two main sections: 'Live Recording' and 'Upload Audio File'. The 'Live Recording' section features a red circular icon and a blue button labeled 'Start Recording'. The 'Upload Audio File' section features a blue icon, a file selection input showing 'ai_voice.mp4', and a blue button labeled 'Analyze File'. Below these sections is a dark blue bar with a loading spinner and the text 'Processing audio with Whisper AI & Gemini LLM... Please wait...'.

Once the file is uploaded, the backend pipeline utilizes Pydub and Wave libraries to process raw audio frames

5. Live Transcription Output

“Transcription: *a is a branch of computer science focused on building software and computer systems capable of performing tasks that traditionally require human intelligence these tasks include learning from experience recognising Complex patterns understanding language and making autonomous decisions instead of following rigid free return code AI models process massive datasets using advanced algorithms to find insights and adopt over time the course of fields of air machine learning deep learning natural language processing and computer vision machine learning is the foundational process where computers analysed data to find patterns and improve a tasks without direct program*

The system handles speech processing using Python's SpeechRecognition library to automatically extract and convert the spoken voice content into clear, structured text lines.

6. Real-time Communication Metrics Display

Student Name:	sri
Evaluation ID:	#3
Timestamp:	2026-07-12 07:07:30
Overall Score:	88/100
Performance Level:	Strong Understanding
Duration:	64.3 sec
Word Count:	96
Words Per Minute (WPM):	89
Similarity Score:	90.72%
Pause Ratio:	17.5%

This display screen showcases computed speech statistics, calculating the runtime duration in seconds, the overall word count, and delivery pacing using Words Per Minute (WPM) metrics.

7.AI Evaluation and Fluency Metrics Analytics


Analysis Result

Conceptual Understanding Score
Strong Understanding

88%


Student Name: sri


File Name: ai_voice.mp4

The text is passed to the Google Gemini 2.5-flash model, which returns a structured JSON payload evaluating conceptual accuracy, qualitative feedback, and predicting exact counts of spoken fluency flaws.

8. Local Database Storage Logs

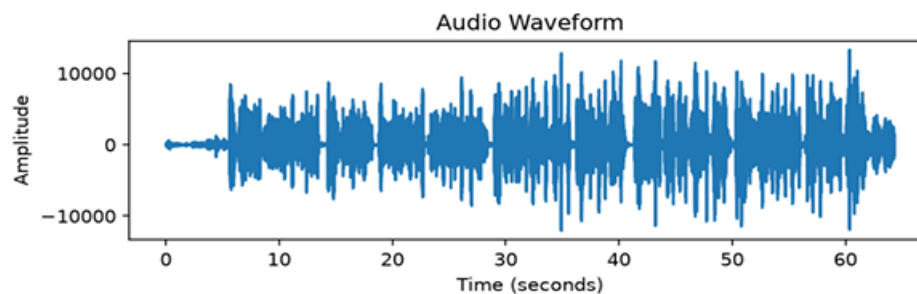
Past Evaluation History						
ID	Student Name	Timestamp	Transcription Extract	Score	Feedback	Action
#4	 sri	2026-07-12 07:17:59	nature refers to the physical world arou...	90%	Strong Understanding	 PDF

Shows the data persistence layer where all analyzed entries—including student names, text transcripts, scores, and JSON filler statistics strings—are safely saved into the fresh_project_records_v2.db table.

9. Downloadable PDF Evaluation Report

Voice-Based Concept Analyser Report

Audio Waveform



Student Name:	sri
Evaluation ID:	#3
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Overall Score:	88/100
Performance Level:	Strong Understanding
Duration:	64.3 sec
Word Count:	96
Words Per Minute (WPM):	89
Similarity Score:	90.72%

The final system output allows a one-click download using ReportLab stream libraries to generate an elegant PDF report containing complete metrics logs, the visual wave graph, and full transcript breakdowns.

Conclusion:

The Voice-Based Concept Understanding Analyser is an AI-powered evaluation platform built with FastAPI that streamlines academic speech assessment and grading. It uses Google's Gemini 2.5-flash model to instantly generate concept scores, filler word statistics, and qualitative feedback from student audio files. The project, featuring automated audio extraction, SQLite logging, and PDF report compilation, proves how targeted AI can boost grading efficiency for educators and students. Looking ahead, the platform can expand by integrating multi-language support, presentation video analysis, and persistent user authentication. By refining model prompts and frontend accessibility, it is well-positioned to evolve into a comprehensive speech management assistant for educational institutions of all sizes.